

Mahdi Zein Al Dine

Beirut, Lebanon | zeinaldinmahdi@gmail.com | +961 71 353 618
[in linkedin.com/in/mahdizd](https://www.linkedin.com/in/mahdizd) | github.com/mahdzd | mahdzd.github.io/mahdizd

Overview

Dedicated Mechatronics Engineering student with research experience in programmable matter, robotics, and embedded systems. Proficient in Python, C++, SolidWorks, and MATLAB, with a Certified SolidWorks Associate (CSWA) certification. Experienced in designing and prototyping engineering systems, providing academic tutoring, and leading student initiatives. Eager to apply strong technical and problem-solving skills to innovative engineering projects.

Education

Lebanese American University (LAU), Byblos, Lebanon Aug. 2023 – May 2027 (Expected)
Bachelor of Engineering (BE) in Mechatronics Engineering
CGPA: 3.87/4.0
Dean's Honor List | Lebanese Baccalaureate Scholarship

Experience

Researcher, Programmable Matter – LAU (Lebanon) - UMLP (France) Jan. 2026 – Present

- Investigating modular reconfigurable robot (MRR) architectures and mechanisms as part of an international research collaboration between LAU and UMLP
- Designing and prototyping a novel multi-output, single-input actuation system for the Datom MRR platform to enhance module functionality while reducing actuator complexity and cost

Academic Tutor – LAU, Beirut Feb. 2024 – Present

- Provided academic support in core Engineering and Mathematics subjects, improving student performance and engagement
- Completed 80+ hours with the official academic success center and 300+ hours of private tutoring

Vice President, Student Council – LAU, Beirut Sep. 2024 – Sep. 2025

- Elected Vice President of the Student Council, representing student interests and contributing to campus-wide decision-making and policy discussions
- Organized and supervised student events and initiatives while coordinating with university administration to address student concerns and improve campus engagement

Honors, Competitions & Certifications

Certifications: Certified SolidWorks Associate (CSWA), Dassault Systèmes (2025)

Awards & Competitions: 3rd Place, Raspberry Pi Competition (IEEE Lebanon Section, 2025); Honorable Mention, Microelectronics Olympiad (LAU, 2026); WRO Future Engineers (Lebanon, 2025)

Selected Projects

WRO Self Driving Car

- Built a 4-wheel Self-Driving Car for the WRO, including advanced control and computer Vision
- [GitHub Repository](#)

VIP+ Solar Panel Cleaning Machine

- Designed a solar-panel cleaning system with the LAU Industrial Hub
- Developed instrumentation using pressure, flow, and TDS sensors with ESP32-based control

Electric Wheelchair - CAD Design & Analysis

- Developed a motorized wheelchair prototype with front-wheel drive, emphasizing user accessibility
- Conducted CAD modeling, motion simulation, and stress analysis using SolidWorks and FEA tools

Skills

Programming: Python, C++, Java **Robotics & Embedded:** Raspberry Pi, ESP32, Arduino **Software:** SolidWorks, AutoCAD, KiCAD, MATLAB, Simulink, SPICE, Git/GitHub, Linux

Languages: Arabic (Native), English (Proficient)